Finite Difference Methods for Partial Differential Equations

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Abstract

Partial Differential Equations (PDE) model the physical phenomena in the universe we live in. To fully understand and interpret those occurring phenomena, PDE must be solved either analytically or numerically. Analytic solution is most needed, unfortunately it is rarely obtained. So, the next best thing is a numerical solution. In this talk, we will discover the so-called Finite Difference Methods (FDM), one of the known methods in numerical analysis. With its given algorithm, a 2D Poisson's equation is examined, simulated in Matlab and compared with its exact solution.

Introduction

Continuous Problem

Let $\Omega = (a, b) \times (a, b)$, Poisson's equation with the Dirichlet condition is written as:

$$\begin{cases} -\Delta u = f(x, y) \text{ in } \Omega \\ u = 0 \text{ on } \partial\Omega \end{cases} \tag{1}$$

where $\Delta u = \dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2}$ is called the Laplacian Operator.

PDE

Definition

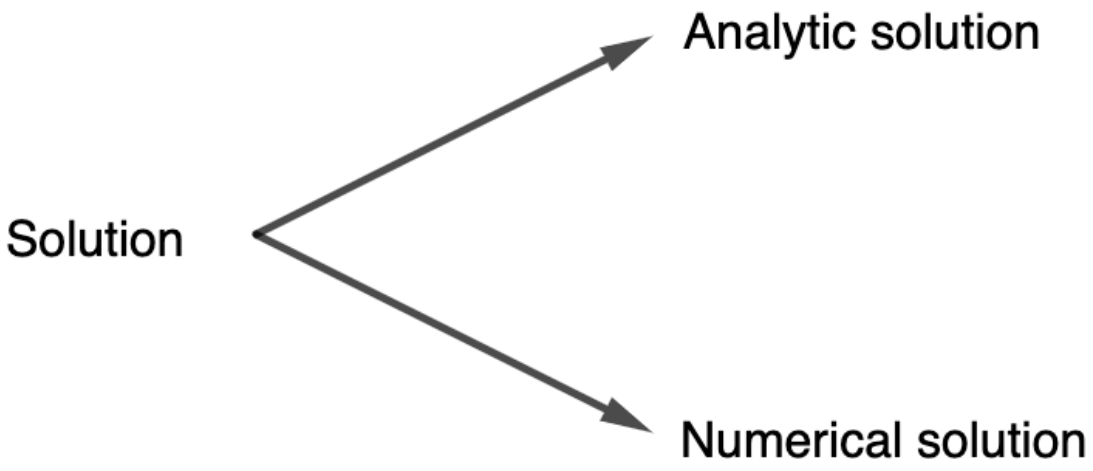
Partial Differential Equations (PDE) is a differential equation containing partial derivatives of the dependent variable (one or more) with more than one independent variable.

Example

1. $-\dfrac{\partial^2 u}{\partial x^2} - \dfrac{\partial^2 u}{\partial y^2} = f(x, y)$: This is the so-called Poisson's equation. The unknown function u is two-variable.
2. $-u''(x) = f(x)$. This is an ODE.

Solutions of PDE

Solving PDE is not like solving algebriac equations. Not only their solutions often unclear, but whether solutions are unique or exist at all are also noble subjects of interest.



Solutions of PDE

Continuous Problem

$$-\Delta u = f(x, y) \xrightarrow{\text{FDM}} 4u_{i,j} - u_{i+1,j} - u_{i-1,j} - u_{i,j+1} - u_{i,j-1} = h^2 f_{i,j}$$

Discretization Problem

Analytic solution

Approximate solution

FDM

Finite difference methods (FDM) are a class of numerical techniques for solving differential equations by approximating derivatives with finite differences. FDM convert PDE into a system of linear equations that can be solved by matrix algebra techniques.

FDM

Algorithms of FDM

1. Mesh Generation: Discretize the domain on which the equation is defined.
2. Problem Discretization: Convert the continuous problem to discrete probblem by replacing, on each node, the derivatives with its approximation.
3. Solve the resulting system of equations.
4. Do the error analysis.

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Approximation to Derivatives

Suppose a function u is 2-time differentiable in the neighborhood of x. For any $h > 0$, we have

$$u(x+h) = u(x) + hu'(x) + \frac{h^2}{2}u''(\xi)$$

where $\xi \in]x, x+h[$.
Or,

$$\left| \frac{u(x+h) - u(x)}{h} - u'(x) \right| \leq Ch, \quad C = \sup_{\xi \in]x, x+h[} \frac{|u''(\xi)|}{2}$$

The derivative u' at point x is approximated by $\dfrac{u(x+h) - u(x)}{h}$, which is known as the *forward difference* approximation.

Approximation to Derivatives

Definition

The approximation U of the derivative u' at point x is of order $p(p > 0)$ if

$$|U - u'| \leq Ch^p, \text{ for some } C > 0$$

Approximation to Derivatives

Suppose that the function u is three times differentiable in the vicinity of x.

$$u(x+h) = u(x) + hu'(x) + \frac{h^2}{2}u''(x) + \frac{h^3}{6}u^{(3)}(\xi^+)$$
$$u(x-h) = u(x) - hu'(x) + \frac{h^2}{2}u''(x) - \frac{h^3}{6}u^{(3)}(\xi^-)$$

Then

$$\frac{u(x+h) - u(x-h)}{2h} = u'(x) + \frac{h^2}{12}(u^{(3)}(\xi^+) + u^{(3)}(\xi^-)) \tag{2}$$

$$\left|\frac{u(x+h) - u(x-h)}{2h} - u'(x)\right| \leq Ch^2, \qquad C = \sup_{\xi \in]x-h, x+h[} \frac{|u^{(3)}(\xi)|}{6}.$$

This defines a second order approximation to u'.

Approximation to Derivatives

Theorem

Suppose that u is a C^4 function on an interval $]x-h, x+h[, h>0$. Then, there exists a constant $C>0$ such that for every $h \in]0, h_0[$ we have:

$$\left| \frac{u(x+h) - 2u(x) + u(x-h)}{h^2} - u''(x) \right| \leq Ch^2 \tag{3}$$

The differential quotient $\dfrac{u(x+h) - 2u(x) + u(x-h)}{h^2}$ is a consistent second-order approximation of the second derivative u'' of u at point x.

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2D Poisson's Equation

Continuous Problem

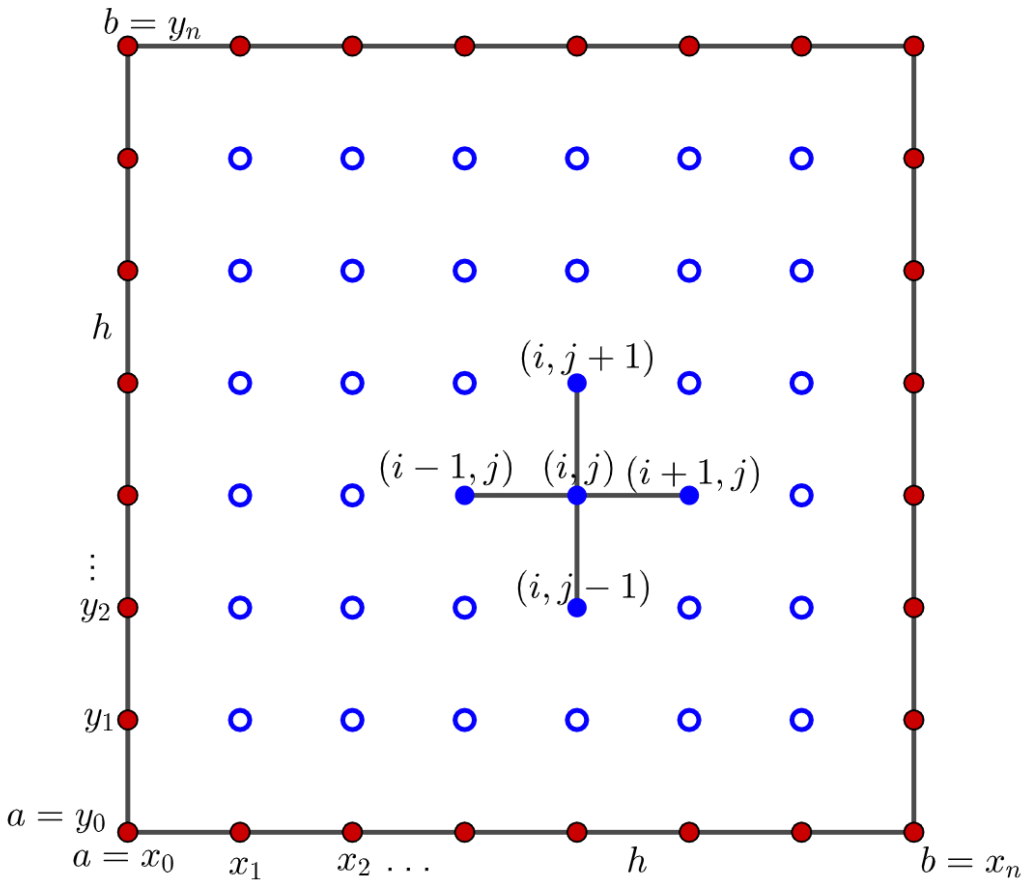
Let $\Omega = (a, b) \times (a, b)$, Poisson's equation with the Dirichlet condition is written as:

$$\begin{cases} -\Delta u = f(x, y) \text{ in } \Omega \\ u = 0 \text{ on } \partial\Omega \end{cases} \tag{4}$$

where $\Delta u = \dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2}$ is called the Laplacian Operator.

Mesh Generation

Take $\Omega = (a, b) \times (a, b)$,



Discrete Problem

At each note on Ω, let

$$u_{i,j} \approx u(x_i, y_j) \text{ and } \Delta_h u_{i,j} \approx \Delta u(x_i, y_j),$$

for $i, j = 0, 1, 2, \ldots, n$. And the Laplacian operator,

$$\begin{aligned} \Delta_h u_{i,j} &= \frac{\partial^2 u_{i,j}}{\partial x^2} + \frac{\partial^2 u_{i,j}}{\partial y^2} \\ &= \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{h^2} + \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{h^2} \end{aligned}$$

Or

$$-\Delta_h u_{i,j} = \frac{1}{h^2}\left(4u_{i,j} - u_{i+1,j} - u_{i-1,j} - u_{i,j+1} - u_{i,j-1}\right), \tag{5}$$

for $i, j = 1, 2, \ldots, n-1$.

Discrete Problem

And the function:

$$f_{i,j} \approx f(x_i, y_j), \qquad \forall i, j = \overline{0, n}$$

Poisson's equation simply is

$$-\Delta_h u_{i,j} = f_{i,j}$$

which is,

$$4u_{i,j} - u_{i+1,j} - u_{i-1,j} - u_{i,j+1} - u_{i,j-1} = h^2 f_{i,j} \tag{6}$$

for $i, j = 1, 2, \ldots, n-1$.
(6) can be written as

$$Au = F, \tag{7}$$

Discrete Problem

$$A = \begin{pmatrix} 4 & -1 & & -1 & & & \\ -1 & \ddots & \ddots & & \ddots & & \\ & \ddots & \ddots & \ddots & & \ddots & \\ -1 & & \ddots & \ddots & \ddots & & -1 \\ & \ddots & & \ddots & \ddots & \ddots & \\ & & \ddots & & \ddots & \ddots & -1 \\ & & & -1 & & -1 & 4 \end{pmatrix}$$

A has the size of $(n-1)^2 \times (n-1)^2$.

Discrete Problem

$$A = \begin{pmatrix} B & -I & 0 & & 0 \\ -I & \ddots & \ddots & \ddots & \\ 0 & \ddots & \ddots & \ddots & 0 \\ & \ddots & \ddots & \ddots & -I \\ 0 & & 0 & -I & B \end{pmatrix},$$

$$B = \begin{pmatrix} 4 & -1 & & & \\ -1 & \ddots & \ddots & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & -1 \\ & & & -1 & 4 \end{pmatrix},$$

0 is zero matrix and I is identity matrix.

Discrete Problem

While vector u and the right-hand side F:

$$u = \begin{pmatrix} u_{1,1} \\ u_{1,2} \\ u_{1,3} \\ \vdots \\ u_{1,n-1} \\ u_{2,1} \\ \vdots \\ u_{n-1,n-1} \end{pmatrix}; \quad F = \begin{pmatrix} h^2 f_{1,1} \\ h^2 f_{1,2} \\ h^2 f_{1,3} \\ \vdots \\ h^2 f_{1,n-1} \\ h^2 f_{2,1} \\ \vdots \\ h^2 f_{n-1,n-1} \end{pmatrix}$$

Simulation

Suppose we want to solve the equation with the Dirichlet boundary condition on the square $\Omega = (0,1)^2$

$$\begin{cases} -\Delta u = 2\pi^2 \sin(\pi x)\sin(\pi y) \\ u(0,y) = u(x,0) = u(1,y) = u(x,1) = 0. \end{cases} \tag{8}$$

The problem (8) has the exact solution $u(x,y) = \sin(\pi x)\sin(\pi y)$.

Simulation

For $n = 4$ then $h = \dfrac{1}{4}$

$$\begin{pmatrix} 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 4 & 0 & 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 4 & -1 & 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & -1 & 4 & -1 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1 & 0 & 0 & 4 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 \end{pmatrix} \cdot \begin{pmatrix} u_{1,1} \\ u_{1,2} \\ u_{1,3} \\ u_{2,1} \\ u_{2,2} \\ u_{2,3} \\ u_{3,1} \\ u_{3,2} \\ u_{3,3} \end{pmatrix} = \begin{pmatrix} 0.61685 \\ 0.87235 \\ 0.61685 \\ 0.87235 \\ 1.23370 \\ 0.87235 \\ 0.61685 \\ 0.87235 \\ 0.61685 \end{pmatrix}$$

Simulation

Solving the system, the approximate solution

$$u = \begin{pmatrix} 0.5265 \\ 0.7446 \\ 0.5265 \\ 0.7446 \\ 1.0530 \\ 0.7446 \\ 0.5265 \\ 0.7446 \\ 0.5265 \end{pmatrix}$$

Simulation

Meshpoint	Exact solution	Approximate solution	Error
$(1/4, 1/4)$	0.5	0.5265	0.0265
$(1/4, 1/2)$	0.7071	0.7446	0.0375
$(1/4, 3/4)$	0.5000	0.5265	0.0265
$(1/2, 1/4)$	0.7071	0.7446	0.0375
$(1/2, 1/2)$	1.0000	1.0530	0.0530
$(1/2, 3/4)$	0.7071	0.7446	0.0375
$(3/4, 1/4)$	0.5000	0.5265	0.0265
$(3/4, 1/2)$	0.7071	0.7446	0.0375
$(3/4, 3/4)$	0.5000	0.5265	0.0265
$\|\|u_{\text{exact}} - u_{\text{app}}\|\|$			0.1061

Table: Comparison between exact and approximate solution at each grid point.

Simulation

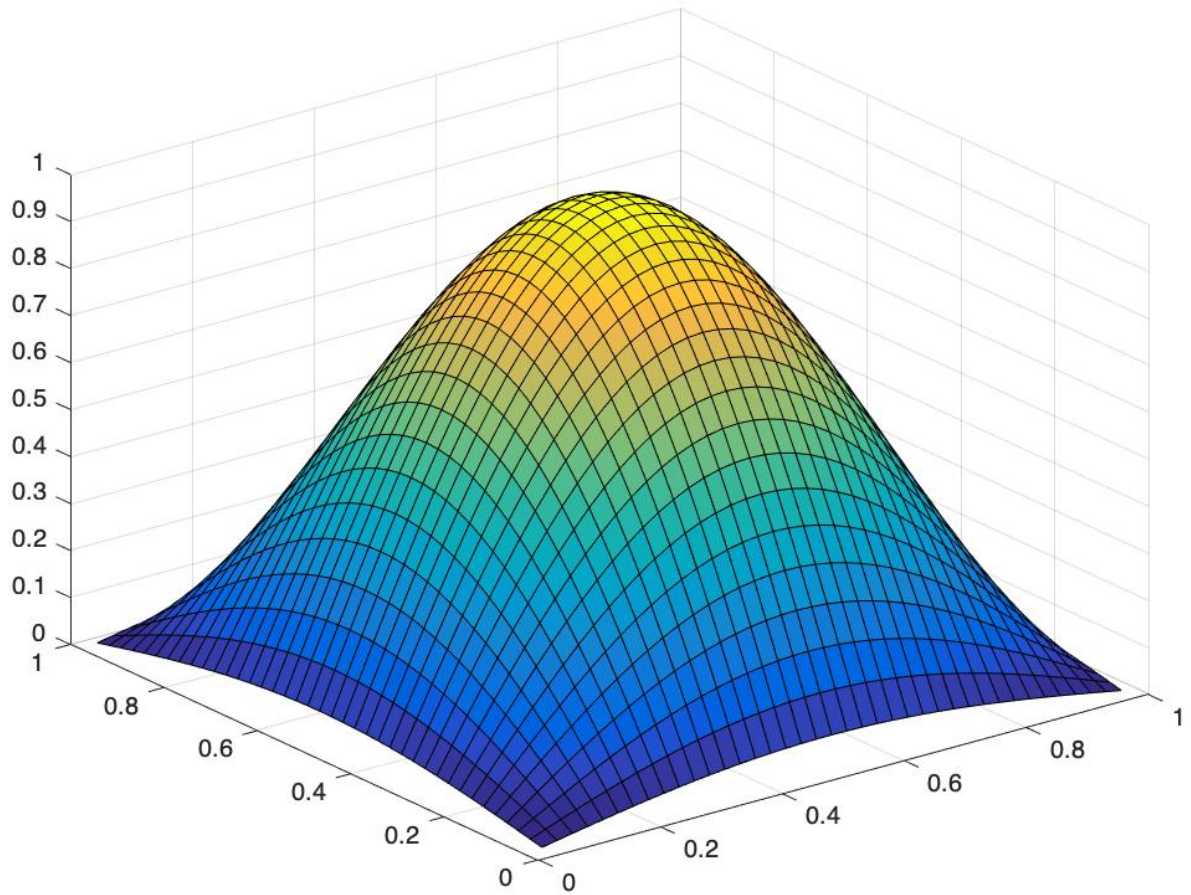


Figure: Exact solution on uniform mesh size for $n = 40$.

Simulation

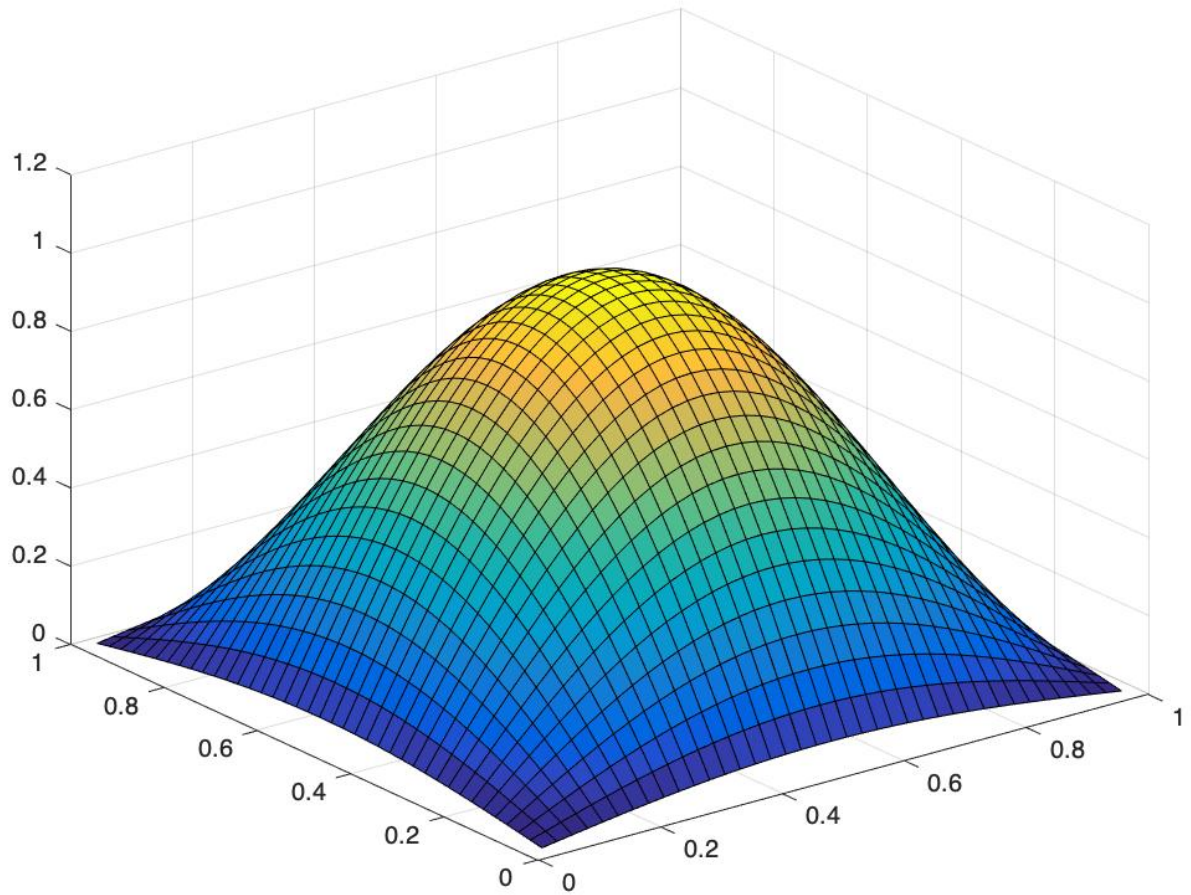


Figure: Approximate solution on uniform mesh size for $n = 40$.

Simulation

No of subintervals n	$\|\|u_{exact} - u_{app}\|\|$
4	0.1061
40	0.0103
90	0.0046
150	0.0027

Table: Comparison between exact and approximate solution with different number of subintervals.

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Conclusion

FDM is very effective in numerically solving the Poisson's problem in 2-dimensional space. The obtained solution is a good approximation to the exact solution.

Math behind the scence

Fundamental Theorem for FDM

$O(h^p)$ local truncation error + stability $\Longrightarrow$ $O(h^p)$ global error.

Math behind the scence

Consistency + stability $\Longrightarrow$ Convergence

- Consistency $||\tau^h|| \leq C_1 h^p$, for $C_1 > 0$.
  Local truncation error:

$$\tau^h = \frac{1}{h^2}(4u_{i,j} - u_{i+1,j} - u_{i-1,j} - u_{i,j+1} - u_{i,j-1}) - f(x_i, y_j) \tag{9}$$

- Stability For any operator A^h, $(A^h)^{-1}$ exists and $||(A^h)^{-1}|| \leq C_2$, for $C_2 > 0$.
- Convergence The error $E^h = u - u^h$ and the equation $A^h E^h = \tau^h$.
  Then if $(A^h)^{-1}$ exists, we have

$$E^h = (A^h)^{-1}\tau^h \Longrightarrow ||E^h|| = ||(A^h)^{-1}\tau^h|| \leq ||(A^h)^{-1}|| \cdot ||\tau^h|| \leq Ch^p \tag{10}$$

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THANK YOU!